JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2018 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
60	100 M	180 Min	MCQ • MSQ • NAT (60 Ques

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2018 • Q1 • ID: JAM_2018_Q1) MCQ +1 M Neg: −0.33

Q. 1 – Q.10 carry one mark each.

Q.1 Which one of the following is TRUE?

(A) $\mathbb{Z}_n$ is cyclic if and only if n is prime

(B) Every proper subgroup of $\mathbb{Z}_n$ is cyclic

(C) Every proper subgroup of S_4 is cyclic

(D) If every proper subgroup of a group is cyclic, then the group is cyclic

Question 2 (JAM 2018 • Q2 • ID: JAM_2018_Q2) MCQ +1 M Neg: −0.33

Q.2 Let $a_n = \frac{b_{n+1}}{b_n}$, where $b_1 = 1$, $b_2 = 1$ and $b_{n+2} = b_n + b_{n+1}$, $n \in \mathbb{N}$. Then $\lim_{n\to\infty} a_n$ is

(A) $\frac{1-\sqrt{5}}{2}$ (B) $\frac{1-\sqrt{3}}{2}$ (C) $\frac{1+\sqrt{3}}{2}$ (D) $\frac{1+\sqrt{5}}{2}$

Question 3 (JAM 2018 • Q3 • ID: JAM_2018_Q3) MCQ +1 M Neg: −0.33

Q.3 If $\{v_1, v_2, v_3\}$ is a linearly independent set of vectors in a vector space over $\mathbb{R}$, then which one of the following sets is also linearly independent?

(A) $\{v_1 + v_2 - v_3,\ 2v_1 + v_2 + 3v_3,\ 5v_1 + 4v_2\}$

(B) $\{v_1 - v_2,\ v_2 - v_3,\ v_3 - v_1\}$

(C) $\{v_1 + v_2 - v_3,\ \ v_2 + v_3 - v_1,\ \ v_3 + v_1 - v_2,\ \ v_1 + v_2 + v_3\}$

(D) $\{v_1 + v_2,\ \ v_2 + 2v_3,\ \ v_3 + 3v_1\}$

Question 4 (JAM 2018 • Q4 • ID: JAM_2018_Q4) MCQ +1 M Neg: −0.33

Q.4 Let a be a positive real number. If f is a continuous and even function defined on the interval $[-a, a]$, then $\int_{-a}^{a} \frac{f(x)}{1+e^x}\,dx$ is equal to

(A) $\int_0^a f(x)\,dx$

(B) $2\int_0^a \frac{f(x)}{1+e^x}\,dx$

(C) $2\int_0^a f(x)\,dx$

(D) $2a\int_0^a \frac{f(x)}{1+e^x}\,dx$

Question 5 (JAM 2018 • Q5 • ID: JAM_2018_Q5) MCQ +1 M Neg: −0.33

Q.5 The tangent plane to the surface $z = \sqrt{x^2 + 3y^2}$ at $(1, 1, 2)$ is given by

(A) $x - 3y + z = 0$

(B) $x + 3y - 2z = 0$

(C) $2x + 4y - 3z = 0$

(D) $3x - 7y + 2z = 0$

Question 6 (JAM 2018 • Q6 • ID: JAM_2018_Q6) MCQ +1 M Neg: −0.33

Q.6 In $\mathbb{R}^3$, the cosine of the acute angle between the surfaces $x^2 + y^2 + z^2 - 9 = 0$ and $z - x^2 - y^2 + 3 = 0$ at the point $(2, 1, 2)$ is

(A) $\frac{8}{5\sqrt{21}}$

(B) $\frac{10}{5\sqrt{21}}$

(C) $\frac{8}{3\sqrt{21}}$

(D) $\frac{10}{3\sqrt{21}}$

Question 7 (JAM 2018 • Q7 • ID: JAM_2018_Q7) MCQ +1 M Neg: −0.33

Q.7 Let $f: \mathbb{R}^3 \to \mathbb{R}$ be a scalar field, $\vec{v}: \mathbb{R}^3 \to \mathbb{R}^3$ be a vector field and let $\vec{a} \in \mathbb{R}^3$ be a constant vector. If $\vec{r}$ represents the position vector $x\hat{\imath} + y\hat{\jmath} + z\hat{k}$, then which one of the following is FALSE?

(A) $curl(f\,\vec{v}) = grad(f) \times \vec{v} + f\,curl(\vec{v})$

(B) $div(grad(f)) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) f$

(C) $curl(\vec{a} \times \vec{r}) = 2\,|\vec{a}|\,\vec{r}$

(D) $div\left(\frac{\vec{r}}{|\vec{r}|^3}\right) = 0$, for $\vec{r} \neq \vec{0}$

Question 8 (JAM 2018 • Q8 • ID: JAM_2018_Q8) MCQ +1 M Neg: −0.33

Q.8 In $\mathbb{R}^2$, the family of trajectories orthogonal to the family of asteroids $x^{2/3} + y^{2/3} = a^{2/3}$ is given by

(A) $x^{4/3} + y^{4/3} = c^{4/3}$

(B) $x^{4/3} - y^{4/3} = c^{4/3}$

(C) $x^{5/3} - y^{5/3} = c^{5/3}$

(D) $x^{2/3} - y^{2/3} = c^{2/3}$

Question 9 (JAM 2018 • Q9 • ID: JAM_2018_Q9) MCQ +1 M Neg: −0.33

Q.9 Consider the vector space V over $\mathbb{R}$ of polynomial functions of degree less than or equal to 3 defined on $\mathbb{R}$. Let $T : V \to V$ be defined by $(Tf)(x) = f(x) - xf'(x)$. Then the rank of T is

(A) 1 (B) 2 (C) 3 (D) 4

Question 10 (JAM 2018 • Q10 • ID: JAM_2018_Q10) **MCQ** **+1 M** **Neg: −0.33**

Q.10 Let $s_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$ for $n \in \mathbb{N}$. Then which one of the following is TRUE for the sequence $\{s_n\}_{n=1}^{\infty}$

(A) $\{s_n\}_{n=1}^{\infty}$ converges in $\mathbb{Q}$

(B) $\{s_n\}_{n=1}^{\infty}$ is a Cauchy sequence but does not converge in $\mathbb{Q}$

(C) the subsequence $\{s_{k^n}\}_{n=1}^{\infty}$ is convergent in $\mathbb{R}$, only when k is even natural number

(D) $\{s_n\}_{n=1}^{\infty}$ is not a Cauchy sequence

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Question 11 (JAM 2018 • Q11 • ID: JAM_2018_Q11) **MCQ** **+2 M** **Neg: −0.67**

Q. 11 – Q. 30 carry two marks each.

Q.11 Let $a_n = \begin{cases} 2 + \frac{(-1)^{\frac{n-1}{2}}}{n} \quad , & if \;\; n \; is \; odd \\ 1 + \frac{1}{2^n} \quad , & if \; n \; is \; even \end{cases} \; , \; n \in \mathbb{N}.$

Then which one of the following is TRUE?

(A) $\sup\ \{a_n \mid n \in \mathbb{N}\} = 3$ and $\inf\{a_n \mid n \in \mathbb{N}\} = 1$

(B) $\liminf\ (a_n) = \limsup\ (a_n) = \frac{3}{2}$

(C) $\sup\{a_n \mid n \in \mathbb{N}\} = 2$ and $\inf\{a_n \mid n \in \mathbb{N}\} = 1$

(D) $\liminf\ (a_n) = 1$ and $\limsup\ (a_n) = 3$

Question 12 (JAM 2018 • Q12 • ID: JAM_2018_Q12) MCQ +2 M Neg: −0.67

Q.12 Let $a, b, c \in \mathbb{R}$. Which of the following values of a, b, c do NOT result in the convergence of the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b (\log_e n)^c} \quad ?$$

(A) $|a| < 1,\ b \in \mathbb{R}, c \in \mathbb{R}$
(B) $a = 1,\ b > 1,\ c \in \mathbb{R}$
(C) $a = 1, b \geq 0, c < 1$
(D) $a = -1,\ b \geq 0, c > 0$

Question 13 (JAM 2018 • Q13 • ID: JAM_2018_Q13) MCQ +2 M Neg: −0.67

Q.13 Let $a_n = n + \frac{1}{n}$, $n \in \mathbb{N}$. Then the sum of the series $\sum_{n=1}^{\infty}(-1)^{n+1} \frac{a_{n+1}}{n!}$ is

(A) $e^{-1} - 1$
(B) e^{-1}
(C) $1 - e^{-1}$
(D) $1 + e^{-1}$

Question 14 (JAM 2018 • Q14 • ID: JAM_2018_Q14) MCQ +2 M Neg: −0.67

Q.14 Let $a_n = \frac{(-1)^n}{\sqrt{1+n}}$ and let $c_n = \sum_{k=0}^{n} a_{n-k} a_k$, where $n \in \mathbb{N} \cup \{0\}$. Then which one of the following is TRUE?

(A) Both $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=1}^{\infty} c_n$ are convergent
(B) $\sum_{n=0}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} c_n$ is not convergent
(C) $\sum_{n=1}^{\infty} c_n$ is convergent but $\sum_{n=0}^{\infty} a_n$ is not convergent
(D) Neither $\sum_{n=0}^{\infty} a_n$ nor $\sum_{n=1}^{\infty} c_n$ is convergent

Question 15 (JAM 2018 • Q15 • ID: JAM_2018_Q15) **MCQ** **+2 M** **Neg: −0.67**

Q.15 Suppose that $f, g : \mathbb{R} \to \mathbb{R}$ are differentiable functions such that f is strictly increasing and g is strictly decreasing. Define $p(x) = f(g(x))$ and $q(x) = g\big(f(x)\big)$, $\forall x \in \mathbb{R}$. Then, for $t > 0$, the sign of $\int_0^t p'(x)\,(\,q'(x) - 3)dx$ is

(A) positive (B) negative (C) dependent on t (D) dependent on f and g

Question 16 (JAM 2018 • Q16 • ID: JAM_2018_Q16) **MCQ** **+2 M** **Neg: −0.67**

Q.16 For $x \in \mathbb{R}$, let $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then which one of the following is FALSE?

(A) $\lim_{x\to 0} \frac{f(x)}{x} = 0$

(B) $\lim_{x\to 0} \frac{f(x)}{x^2} = 0$

(C) $\frac{f(x)}{x^2}$ has infinitely many maxima and minima on the interval $(0,1)$

(D) $\frac{f(x)}{x^4}$ is continuous at $x = 0$ but not differentiable at $x = 0$

Question 17 (JAM 2018 • Q17 • ID: JAM_2018_Q17) **MCQ** **+2 M** **Neg: −0.67**

Q.17 Let $f(x,y) = \begin{cases} \frac{xy}{(x^2+y^2)^\alpha}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

Then which one of the following is TRUE for f at the point $(0,0)$?

(A) For $\alpha = 1$, f is continuous but not differentiable

(B) For $\alpha = \frac{1}{2}$, f is continuous and differentiable

(C) For $\alpha = \frac{1}{4}$, f is continuous and differentiable

(D) For $\alpha = \frac{3}{4}$, f is neither continuous nor differentiable

Question 18 (JAM 2018 • Q18 • ID: JAM_2018_Q18) MCQ +2 M Neg: −0.67

Q.18 Let $a, b \in \mathbb{R}$ and let $f: \mathbb{R} \to \mathbb{R}$ be a thrice differentiable function. If $z = e^u f(v)$, where $u = ax + by$ and $v = ax - by$, then which one of the following is TRUE?

(A) $b^2 z_{xx} - a^2 z_{yy} = 4a^2 b^2 e^u f'(v)$

(B) $b^2 z_{xx} - a^2 z_{yy} = -4e^u f'(v)$

(C) $bz_x + az_y = abz$

(D) $bz_x + az_y = -abz$

Question 19 (JAM 2018 • Q19 • ID: JAM_2018_Q19) MCQ +2 M Neg: −0.67

Q.19 Consider the region D in the yz plane bounded by the line $y = \frac{1}{2}$ and the curve $y^2 + z^2 = 1$, where $y \geq 0$. If the region D is revolved about the z-axis in $\mathbb{R}^3$, then the volume of the resulting solid is

(A) $\frac{\pi}{\sqrt{3}}$

(B) $\frac{2\pi}{\sqrt{3}}$

(C) $\frac{\pi\sqrt{3}}{2}$

(D) $\pi\sqrt{3}$

Question 20 (JAM 2018 • Q20 • ID: JAM_2018_Q20) MCQ +2 M Neg: −0.67

Q.20 If $\vec{F}(x, y) = (3x - 8y)\hat{\imath} + (4y - 6xy)\hat{\jmath}$ for $(x, y) \in \mathbb{R}^2$, then $\oint_C \vec{F} \cdot d\,\vec{r}$, where C is the boundary of the triangular region bounded by the lines $x = 0, y = 0$ and $x + y = 1$ oriented in the anti-clockwise direction, is

(A) $\frac{5}{2}$

(B) 3

(C) 4

(D) 5

Question 21 (JAM 2018 • Q21 • ID: JAM_2018_Q21) **MCQ** **+2 M** **Neg: −0.67**

Q.21 Let U, V and W be finite dimensional real vector spaces, $T: U \to V$, $S: V \to W$ and $P: W \to U$ be linear transformations. If range (ST) = nullspace (P), nullspace (ST) = range (P) and rank (T) = rank (S), then which one of the following is TRUE?

(A) nullity of T = nullity of S

(B) dimension of $U \neq$ dimension of W

(C) If dimension of $V = 3$, dimension of $U = 4$, then P is not identically zero

(D) If dimension of $V = 4$, dimension of $U = 3$ and T is one-one, then P is identically zero

Question 22 (JAM 2018 • Q22 • ID: JAM_2018_Q22) **MCQ** **+2 M** **Neg: −0.67**

Q.22 Let $y(x)$ be the solution of the differential equation $\frac{dy}{dx} + y = f(x)$, for $x \geq 0$, $y(0) = 0$, where

$f(x) = \begin{cases} 2, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$. Then $y(x) =$

(A) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2(e - 1)e^{-x}$ when $x \geq 1$

(B) $2(1 - e^{-x})$ when $0 \leq x < 1$ and 0 when $x \geq 1$

(C) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2(1 - e^{-1})e^{-x}$ when $x \geq 1$

(D) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2\, e^{1-x}$ when $x \geq 1$

Question 23 (JAM 2018 • Q23 • ID: JAM_2018_Q23) **MCQ** **+2 M** **Neg: −0.67**

Q.23 An integrating factor of the differential equation $\left(y + \frac{1}{3} y^3 + \frac{1}{2} x^2\right) dx + \frac{1}{4}(x + x\, y^2)\, dy = 0$ is

(A) x^2 (B) $3 \log_e x$ (C) x^3 (D) $2 \log_e x$

Question 24 (JAM 2018 • Q24 • ID: JAM_2018_Q24) MCQ +2 M Neg: −0.67

Q.24 A particular integral of the differential equation $y'' + 3y' + 2y = e^{e^x}$ is

(A) $e^{e^x}e^{-x}$ (B) $e^{e^x}e^{-2x}$ (C) $e^{e^x}e^{2x}$ (D) $e^{e^x}e^{x}$

Question 25 (JAM 2018 • Q25 • ID: JAM_2018_Q25) MCQ +2 M Neg: −0.67

Q.25 Let G be a group satisfying the property that $f : G \to \mathbb{Z}_{221}$ is a homomorphism implies $f(g) = 0,\ \forall g \in G$. Then a possible group G is

(A) $\mathbb{Z}_{21}$ (B) $\mathbb{Z}_{51}$ (C) $\mathbb{Z}_{91}$ (D) $\mathbb{Z}_{119}$

Question 26 (JAM 2018 • Q26 • ID: JAM_2018_Q26) MCQ +2 M Neg: −0.67

Q.26 Let H be the quotient group $\mathbb{Q}/\mathbb{Z}$. Consider the following statements.

I. Every cyclic subgroup of H is finite.

II. Every finite cyclic group is isomorphic to a subgroup of H.

Which one of the following holds?

(A) I is TRUE but II is FALSE (B) II is TRUE but I is FALSE

(C) both I and II are TRUE (D) neither I nor II is TRUE

Question 27 (JAM 2018 • Q27 • ID: JAM_2018_Q27) MCQ +2 M Neg: −0.67

Q.27 Let I denote the 4×4 identity matrix. If the roots of the characteristic polynomial of a 4×4 matrix M are $\pm\sqrt{\frac{1\pm\sqrt{5}}{2}}$, then $M^8 =$

(A) $I + M^2$ (B) $2I + M^2$ (C) $2I + 3M^2$ (D) $3I + 2M^2$

Question 28 (JAM 2018 • Q28 • ID: JAM_2018_Q28) **MCQ** **+2 M** **Neg: −0.67**

Q.28 Consider the group $\mathbb{Z}^2 = \{(a,b) \mid a,b \in \mathbb{Z}\}$ under component-wise addition. Then which of the following is a subgroup of $\mathbb{Z}^2$?

(A) $\{(a,b) \in \mathbb{Z}^2 \mid ab = 0\}$

(B) $\{(a,b) \in \mathbb{Z}^2 \mid 3a + 2b = 15\}$

(C) $\{(a,b) \in \mathbb{Z}^2 \mid 7 \text{ divides } ab\}$

(D) $\{(a,b) \in \mathbb{Z}^2 \mid 2 \text{ divides } a \text{ and } 3 \text{ divides } b\}$

Question 29 (JAM 2018 • Q29 • ID: JAM_2018_Q29) **MCQ** **+2 M** **Neg: −0.67**

Q.29 Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function and let J be a bounded open interval in $\mathbb{R}$. Define

$$W(f,J) = \sup\,\{f(x) \mid x \in J\} \;-\; \inf\,\{f(x) \mid x \in J\}\ .$$

Which one of the following is FALSE?

(A) $W(f,J_1) \leq W(f,J_2)$ if $J_1 \subset J_2$

(B) If f is a bounded function in J and $J \supset J_1 \supset J_2 \cdots \supset J_n \supset \cdots$ such that the length of the interval J_n tends to 0 as $n \to \infty$, then $\lim\limits_{n\to\infty} W(f,J_n) = 0$

(C) If f is discontinuous at a point $a \in J$, then $W(f,J) \neq 0$

(D) If f is continuous at a point $a \in J$, then for any given $\epsilon > 0$ there exists an interval $I \subset J$ such that $W(f,I) < \epsilon$

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Question 30 (JAM 2018 • Q30 • ID: JAM_2018_Q30) MCQ +2 M Neg: −0.67

Q.30 For $x > \frac{-1}{2}$, let $f_1(x) = \frac{2x}{1+2x}$, $f_2(x) = \log_e(1 + 2x)$ and $f_3(x) = 2x$. Then which one of the following is TRUE?

(A) $f_3(x) < f_2(x) < f_1(x)$ for $0 < x < \frac{\sqrt{3}}{2}$

(B) $f_1(x) < f_3(x) < f_2(x)$ for $x > 0$

(C) $f_1(x) + f_2(x) < \frac{f_3(x)}{2}$ for $x > \frac{\sqrt{3}}{2}$

(D) $f_2(x) < f_1(x) < f_3(x)$ for $x > 0$

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each

Question 31 (JAM 2018 • Q31 • ID: JAM_2018_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 – Q. 40 carry two marks each.

Q.31 Let $f: \mathbb{R}\setminus\{0\} \to \mathbb{R}$ be defined by $f(x) = x + \frac{1}{x^3}$. On which of the following interval(s) is f one-one?

(A) $(-\infty, -1)$ (B) $(0, 1)$ (C) $(0, 2)$ (D) $(0, \infty)$

Question 32 (JAM 2018 • Q32 • ID: JAM_2018_Q32) MSQ +2 M Neg: Nil (0)

Q.32 The solution(s) of the differential equation $\frac{dy}{dx} = (\sin 2x)\, y^{1/3}$ satisfying $y(0) = 0$ is (are)

(A) $y(x) = 0$

(B) $y(x) = -\sqrt{\frac{8}{27}}\, \sin^3 x$

(C) $y(x) = \sqrt{\frac{8}{27}}\, \sin^3 x$

(D) $y(x) = \sqrt{\frac{8}{27}}\, \cos^3 x$

Question 33 (JAM 2018 • Q33 • ID: JAM_2018_Q33) MSQ +2 M Neg: Nil (0)

Q.33 Suppose f, g, h are permutations of the set $\{\alpha, \beta, \gamma, \delta\}$, where

f interchanges α and β but fixes γ and δ,

g interchanges β and γ but fixes α and δ,

h interchanges γ and δ but fixes α and β.

Which of the following permutations interchange(s) α and δ but fix(es) β and γ?

(A) $f \circ g \circ h \circ g \circ f$ (B) $g \circ h \circ f \circ h \circ g$ (C) $g \circ f \circ h \circ f \circ g$ (D) $h \circ g \circ f \circ g \circ h$

Question 34 (JAM 2018 • Q34 • ID: JAM_2018_Q34) MSQ +2 M Neg: Nil (0)

Q.34 Let P and Q be two non-empty disjoint subsets of $\mathbb{R}$. Which of the following is (are) FALSE?

(A) If P and Q are compact, then $P \cup Q$ is also compact

(B) If P and Q are not connected, then $P \cup Q$ is also not connected

(C) If $P \cup Q$ and P are closed, then Q is closed

(D) If $P \cup Q$ and P are open, then Q is open

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Question 35 (JAM 2018 • Q35 • ID: JAM_2018_Q35) MSQ +2 M Neg: Nil (0)

Q.35 Let $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ denote the group of non-zero complex numbers under multiplication. Suppose $Y_n = \{z \in \mathbb{C} \mid z^n = 1\}$, $n \in \mathbb{N}$. Which of the following is (are) subgroup(s) of $\mathbb{C}^*$?

(A) $\bigcup_{n=1}^{100} Y_n$ (B) $\bigcup_{n=1}^{\infty} Y_{2^n}$ (C) $\bigcup_{n=100}^{\infty} Y_n$ (D) $\bigcup_{n=1}^{\infty} Y_n$

Question 36 (JAM 2018 • Q36 • ID: JAM_2018_Q36) MSQ +2 M Neg: Nil (0)

Q.36 Suppose $\alpha, \beta, \gamma \in \mathbb{R}$. Consider the following system of linear equations.
$x + y + z = \alpha,\ x + \beta y + z = \gamma,\ \ x + y + \alpha z = \beta$. If this system has at least one solution, then which of the following statements is (are) TRUE?

(A) If $\alpha = 1$ then $\gamma = 1$

(B) If $\beta = 1$ then $\gamma = \alpha$

(C) If $\beta \neq 1$ then $\alpha = 1$

(D) If $\gamma = 1$ then $\alpha = 1$

Question 37 (JAM 2018 • Q37 • ID: JAM_2018_Q37) MSQ +2 M Neg: Nil (0)

Q.37 Let $m, n \in \mathbb{N},\ \ m < n,\ \ \ P \in M_{n\times m}(\mathbb{R}),\ \ Q \in M_{m\times n}(\mathbb{R})$. Then which of the following is (are) NOT possible?

(A) $rank\ (PQ) = n$

(B) $rank\ (QP) = m$

(C) $rank\ (PQ) = m$

(D) $rank\ (QP) = \left\lceil \frac{m+n}{2} \right\rceil$, the smallest integer larger than or equal to $\frac{m+n}{2}$

Question 38 (JAM 2018 • Q38 • ID: JAM_2018_Q38) MSQ +2 M Neg: Nil (0)

Q.38 If $\vec{F}(x,y,z) = (2x + 3yz)\hat{\imath} + (3xz + 2y)\hat{\jmath} + (3xy + 2z)\hat{k}$ for $(x,y,z) \in \mathbb{R}^3$, then which among the following is (are) TRUE?

(A) $\nabla \times \vec{F} = \vec{0}$

(B) $\oint_C\ \vec{F}\cdot d\vec{r} =\ 0$ along any simple closed curve C

(C) There exists a scalar function $\phi : \mathbb{R}^3 \to \mathbb{R}$ such that $\nabla \cdot \vec{F} =\ \phi_{xx} + \phi_{yy} + \phi_{zz}$

(D) $\nabla \cdot \vec{F} = 0$

Question 39 (JAM 2018 • Q39 • ID: JAM_2018_Q39) MSQ +2 M Neg: Nil (0)

Q.39 Which of the following subsets of $\mathbb{R}$ is (are) connected?

(A) $\{x \in \mathbb{R} \mid x^2 + x > 4\}$
(B) $\{x \in \mathbb{R} \mid x^2 + x < 4\}$
(C) $\{x \in \mathbb{R} \mid |x| < |x-4|\}$
(D) $\{x \in \mathbb{R} \mid |x| > |x-4|\}$

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Question 40 (JAM 2018 • Q40 • ID: JAM_2018_Q40) MSQ +2 M Neg: Nil (0)

Q.40 Let S be a subset of $\mathbb{R}$ such that 2018 is an interior point of S. Which of the following is (are) TRUE?

(A) S contains an interval
(B) There is a sequence in S which does not converge to 2018
(C) There is an element $y \in S$, $y \neq 2018$ such that y is also an interior point of S
(D) There is a point $z \in S$, such that $|z - 2018| = 0.002018$

SECTION – C
NUMERICAL ANSWER TYPE (NAT)

Q. 41 – Q. 50 carry one mark each.

Question 41 (JAM 2018 • Q41 • ID: JAM_2018_Q41) NAT +1 M Neg: Nil (0)

Q. 41 – Q. 50 carry one mark each.

Q.41 The order of the element $(1\ 2\ 3)\,(2\ 4\ 5)\,(4\ 5\ 6)$ in the group S_6 is ________________

Question 42 (JAM 2018 • Q42 • ID: JAM_2018_Q42) NAT +1 M Neg: Nil (0)

Q.42 Let $\phi(x, y, z) = 3y^2 + 3yz$ for $(x, y, z) \in \mathbb{R}^3$. Then the absolute value of the directional derivative of ϕ in the direction of the line $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z}{-2}$, at the point $(1, -2, 1)$ is ______________

Question 43 (JAM 2018 • Q43 • ID: JAM_2018_Q43) NAT +1 M Neg: Nil (0)

Q.43 Let $f(x) = \sum_{n=0}^{\infty}(-1)^n x(x-1)^n$ for $0 < x < 2$. Then the value of $f\left(\frac{\pi}{4}\right)$ is ____________

Question 44 (JAM 2018 • Q44 • ID: JAM_2018_Q44) NAT +1 M Neg: Nil (0)

Q.44 Let $f: \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x,y) = \begin{cases} \dfrac{x^2 y\,(x-y)}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Then $\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) - \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$ at the point $(0,0)$ is ____________

Question 45 (JAM 2018 • Q45 • ID: JAM_2018_Q45) NAT +1 M Neg: Nil (0)

Q.45 Let $f(x,y) = \sqrt{x^3 y}\,\sin\left(\frac{\pi}{2}\,e^{\left(\frac{y}{x}-1\right)}\right) + xy\,\cos\left(\frac{\pi}{3}\,e^{\left(\frac{x}{y}-1\right)}\right)$ for $(x,y) \in \mathbb{R}^2,\ x > 0,\ y > 0$.

Then $f_x(1,1) + f_y(1,1) =$ _______________

Question 46 (JAM 2018 • Q46 • ID: JAM_2018_Q46) NAT +1 M Neg: Nil (0)

Q.46 Let $f : [0, \infty) \to [0, \infty)$ be continuous on $[0, \infty)$ and differentiable on $(0, \infty)$. If $f(x) = \int_0^x \sqrt{f(t)}\, dt$, then $f(6) =$ ___________

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Question 47 (JAM 2018 • Q47 • ID: JAM_2018_Q47) NAT +1 M Neg: Nil (0)

Q.47 Let $a_n = \frac{(1+(-1)^n)}{2^n} + \frac{(1+(-1)^{n-1})}{3^n}$. Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ about $x = 0$ is __________

Question 48 (JAM 2018 • Q48 • ID: JAM_2018_Q48) NAT +1 M Neg: Nil (0)

Q.48 Let A_6 be the group of even permutations of 6 distinct symbols. Then the number of elements of order 6 in A_6 is __________________

Question 49 (JAM 2018 • Q49 • ID: JAM_2018_Q49) NAT +1 M Neg: Nil (0)

Q.49 Let W_1 be the real vector space of all 5×2 matrices such that the sum of the entries in each row is zero. Let W_2 be the real vector space of all 5×2 matrices such that the sum of the entries in each column is zero. Then the dimension of the space $W_1 \cap W_2$ is ______________

Question 50 (JAM 2018 • Q50 • ID: JAM_2018_Q50) NAT +1 M Neg: Nil (0)

Q.50 The coefficient of x^4 in the power series expansion of $e^{\sin x}$ about $x = 0$ is ____________ (correct up to three decimal places).

Q. 51 – Q. 60 carry two marks each.

Question 51 (JAM 2018 • Q51 • ID: JAM_2018_Q51) NAT +2 M Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.

Q.51 Let $a_k = (-1)^{k-1}$, $s_n = a_1 + a_2 + \cdots + a_n$ and $\sigma_n = (s_1 + s_2 + \cdots + s_n)/n$, where $k, n \in \mathbb{N}$. Then $\lim_{n\to\infty} \sigma_n$ is _______________ (correct up to one decimal place).

Question 52 (JAM 2018 • Q52 • ID: JAM_2018_Q52) NAT +2 M Neg: Nil (0)

Q.52 Let $f: \mathbb{R} \to \mathbb{R}$ be such that f'' is continuous on $\mathbb{R}$ and $f(0) = 1$, $f'(0) = 0$ and $f''(0) = -1$.

Then $\lim_{x\to\infty} \left(f\left(\sqrt{\frac{2}{x}}\right)\right)^x$ is ________________ (correct up to three decimal places).

Question 53 (JAM 2018 • Q53 • ID: JAM_2018_Q53) NAT +2 M Neg: Nil (0)

Q.53 Suppose x, y, z are positive real numbers such that $x + 2y + 3z = 1$. If M is the maximum value of xyz^2, then the value of $\frac{1}{M}$ is ________________

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Question 54 (JAM 2018 • Q54 • ID: JAM_2018_Q54) NAT +2 M Neg: Nil (0)

Q.54 If the volume of the solid in $\mathbb{R}^3$ bounded by the surfaces

$$x = -1, \quad x = 1, \quad y = -1, \quad y = 1, \quad z = 2, \quad y^2 + z^2 = 2$$

is $\alpha - \pi$, then $\alpha =$ _______________

Question 55 (JAM 2018 • Q55 • ID: JAM_2018_Q55) NAT +2 M Neg: Nil (0)

Q.55 If $\alpha = \int_{\pi/6}^{\pi/3} \frac{\sin t + \cos t}{\sqrt{\sin 2t}}\, dt$, then the value of $\left(2\sin\frac{\alpha}{2} + 1\right)^2$ is _____________

Question 56 (JAM 2018 • Q56 • ID: JAM_2018_Q56) NAT +2 M Neg: Nil (0)

Q.56 The value of the integral

$$\int_0^1 \int_x^1 y^4 e^{xy^2}\, dy\, dx$$

is _____________ (correct up to three decimal places).

Question 57 (JAM 2018 • Q57 • ID: JAM_2018_Q57) NAT +2 M Neg: Nil (0)

Q.57 Suppose $Q \in M_{3\times 3}(\mathbb{R})$ is a matrix of rank 2. Let $T: M_{3\times 3}(\mathbb{R}) \to M_{3\times 3}(\mathbb{R})$ be the linear transformation defined by $T(P) = QP$. Then the rank of T is ______________

Question 58 (JAM 2018 • Q58 • ID: JAM_2018_Q58) NAT +2 M Neg: Nil (0)

Q.58 The area of the parametrized surface

$S = \{((2+\cos u)\cos v,\ (2+\cos u)\sin v,\ \sin u) \in \mathbb{R}^3 \mid 0 \le u \le \frac{\pi}{2},\ 0 \le v \le \frac{\pi}{2}\}$

is ____________ (correct up to two decimal places).

Question 59 (JAM 2018 • Q59 • ID: JAM_2018_Q59) NAT +2 M Neg: Nil (0)

Q.59 If $x(t)$ is the solution to the differential equation $\frac{dx}{dt} = x^2t^3 + xt$, for $t > 0$, satisfying $x(0) = 1$, then the value of $x(\sqrt{2})$ is _____________ (correct up to two decimal places).

Question 60 (JAM 2018 • Q60 • ID: JAM_2018_Q60) NAT +2 M Neg: Nil (0)

Q.60 If $y(x) = v(x)\sec x$ is the solution of $y'' - (2\tan x)\,y' + 5y = 0$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$, satisfying $y(0) = 0$ and $y'(0) = \sqrt{6}$, then $v\left(\frac{\pi}{6\sqrt{6}}\right)$ is _______________ (correct up to two decimal places).

END OF THE QUESTION PAPER

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OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2018 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+1	B
2	MCQ	+1	D
3	MCQ	+1	D
4	MCQ	+1	A
5	MCQ	+1	B
6	MCQ	+1	C
7	MCQ	+1	C
8	MCQ	+1	B
9	MCQ	+1	C
10	MCQ	+1	B
11	MCQ	+2	A
12	MCQ	+2	C
13	MCQ	+2	D
14	MCQ	+2	B
15	MCQ	+2	A
16	MCQ	+2	D
17	MCQ	+2	C
18	MCQ	+2	A
19	MCQ	+2	C
20	MCQ	+2	B
21	MCQ	+2	C
22	MCQ	+2	A
23	MCQ	+2	C
24	MCQ	+2	B
25	MCQ	+2	A
26	MCQ	+2	C
27	MCQ	+2	C
28	MCQ	+2	D
29	MCQ	+2	B
30	MCQ	+2	Marks to All
31	MSQ	+2	B
32	MSQ	+2	A;B;C
33	MSQ	+2	A;D
34	MSQ	+2	B;C;D
35	MSQ	+2	B;C;D
36	MSQ	+2	A;B
37	MSQ	+2	A;D
38	MSQ	+2	A;B;C
39	MSQ	+2	B;C;D
40	MSQ	+2	A;B;C
41	NAT	+1	4 to 4
42	NAT	+1	6.5 to 7.5
43	NAT	+1	1 to 1
44	NAT	+1	1 to 1
45	NAT	+1	3 to 3
46	NAT	+1	9 to 9
47	NAT	+1	2 to 2
48	NAT	+1	0 to 0
49	NAT	+1	4 to 4
50	NAT	+1	−0.130 to −0.120
51	NAT	+2	0.4 to 0.6
52	NAT	+2	0.350 to 0.380
53	NAT	+2	1140 to 1160
54	NAT	+2	5.99 to 6.01
55	NAT	+2	2.9 to 3.1
56	NAT	+2	0.230 to 0.250
57	NAT	+2	6 to 6
58	NAT	+2	6.30 to 6.70
59	NAT	+2	−2.80 to −2.70
60	NAT	+2	0.5 to 0.5

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.